

DISCLAIMER: *This exercise is strictly methodological in nature. It is not intended here to dispute and/or replace the official dating of economic cycles carried out by CODACE (FGV/IBRE), which remains the institutional reference. The objective is to bring to light the dynamics of state transitions from the point of view of stochastic modeling.*

Beyond the Average: What Markov Chains Can Teach Us About Brazilian Inertia

One of the fundamental characteristics of the Brazilian economy is its profound non-linearity. In Brazil, this behavior is practically the *status quo*: conjunctural interferences—treated here as noise—have the power to abruptly alter the momentary trajectory of activity. This noise operates in jumps and ruptures that simple linear models often fail to capture. For the economic analyst, the central question is not just "how much the GDP will grow," but "in what state of nature are we living". This distinction is vital for business strategies and public policies to be precisely coordinated by decision-makers.

Methodologies: The Deterministic vs. The Stochastic

Business Cycle Dating Committee (CODACE)

The [Business Cycle Dating Committee](#) (CODACE) is responsible for establishing the official chronology of recessions and expansions in the Brazilian economy. To this end, it adopts the deterministic *Bry-Boschan* methodology, originally developed for the **NBER** in the United States, which identifies peak and trough points in economic cycles based on the analysis of the duration, amplitude, and spread of variations in activity, combined with the expertise of renowned specialists in the economic field. The committee considers a broad set of variables, such as quarterly GDP, industrial production, consumption, investment, and the labor market, avoiding reliance solely on simplified rules, such as two consecutive quarters of negative GDP, and offering a more accurate view of the country's economic dynamics.

Markov Chain (Markov-Switching)

The Markov Chain is a stochastic model that describes probabilistic systems in which the future state depends only on the present state, and not on the sequence of past states. This property, called Markovian, makes the process “memoryless” and allows the evolution of phenomena over time to be represented through a transition matrix, which contains the probabilities of moving from one state to another. From there, the Markov chain estimates the transition probabilities between these regimes, offering a probabilistic view of economic cycles and the dynamics of activity. In a very summarized way, it would be the following question: *What is the real chance of an economy changing to another level in the next period?*

In the proposed exercise, which used the Brazilian Gross Domestic Product and followed the same binary concept as CODACE—expansion and recession—the regimes present the following characteristics:

- **Recession Regime:** Moments of contraction where negative inertia (p_{11}) requires structural change for a return to normalcy.

- **Expansion Regime:** Periods of traction where the probability of remaining in the current state (p_{22}) is the driver of growth.

The Fragility of Brazilian Expansion

When analyzing the Transition Matrix and the model parameters, one piece of data stands out: residual volatility in the expansionary regime frequently exceeds the growth rate itself.

Transition Matrix

Regimes	To Recession	To Expansion
From Recession	0.5328	0.0348
From Expansion	0.4672	0.9652

This indicates that, even in periods of growth, *statistical noise* and quarterly shocks are so intense that the expansion signal becomes fragile. In Brazil, expansion is a volatile state; the force of inertia and vulnerability to external shocks remain the main determinants of our long-term trajectory.

The Anatomy of Regimes: The Weight of the Coefficients

The data demonstrate a severe asymmetry: while the recession regime imposes an average contraction of **2.24%** per quarter, the so-called "expansion" presents an anemic pace of only **0.79%**, showing that our recovery breath is marginal compared to the impact of crises.

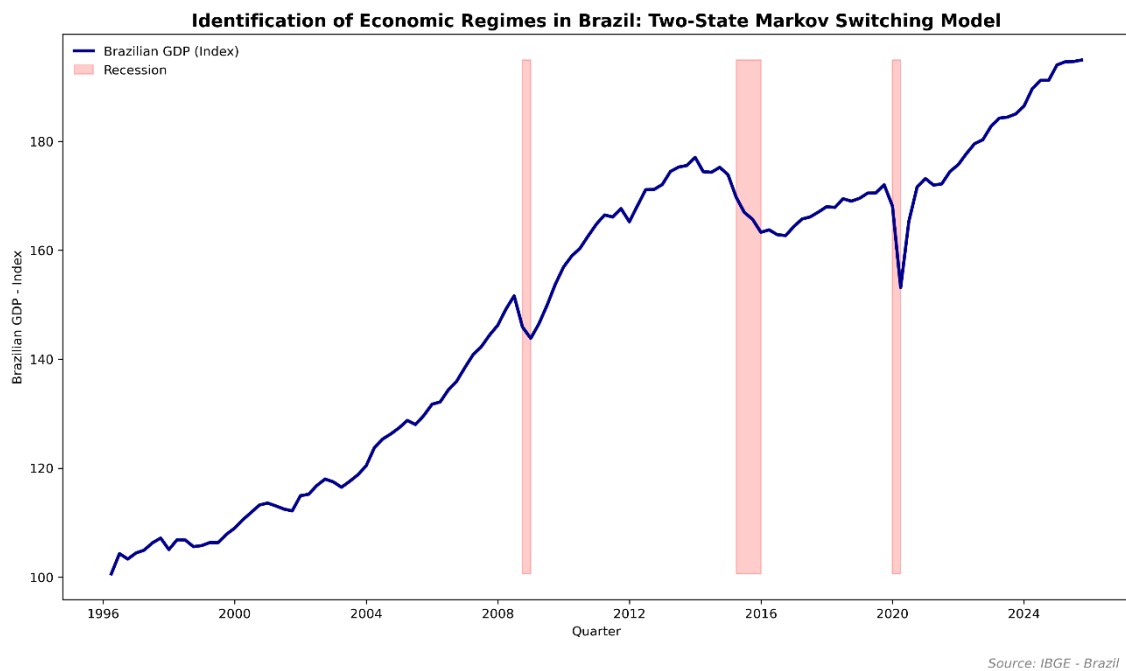
Parameter Coefficients

	Coefficient	Standard Error	Z-Score	P-Value
Probability from Recession to Recession	0.5328	0.2045	2.6058	0.0092
Probability from Expansion to Recession	0.0348	0.0206	1.6935	0.0904
Constant _{Recession}	-2.2416	0.4870	-4.6026	0.0000
Constant _{Expansion}	0.7939	0.1072	7.4073	0.0000
σ^2	1.0929	0.1547	7.0658	0.0000

This inertia is reinforced by the probability of remaining in recession (p_{11}), which reaches **53.3%**, contrasting with the low probability of **3.5%** for a direct transition from bonanza to collapse. The Sigma2 coefficient of **1.1** is the element that quantifies this instability; by representing a residual variance that exceeds the growth rate of the expansionary regime itself, it indicates that quarterly shocks and statistical noise frequently overcome the growth signal. Thus, Brazilian expansion becomes a fragile and volatile state, where the force of inertia remains the main determinant of the long-term trajectory.

Chronology of Identified Regimes

In the graph below, which covers the 1st quarter of 1996 to the 4th quarter of 2025, the result of this methodology is presented. It can be seen that the Markov Chain identified three periods of recession:



The model accurately captured the breaking points of our recent history:

- **4th Quarter of 2008 to 1st Quarter of 2009:** The impact of the Global Financial Crisis (Subprime). The model detects the sudden contraction in exports and credit, converging exactly with the CODACE dating.
- **2nd Quarter of 2015 to 1st Quarter of 2016:** The peak of the political crisis and fiscal imbalance. Although CODACE identifies a more extensive period (starting in 2014), the Markov model focuses on the *core* of the recession, where the regime change was statistically undeniable.
- **1st Quarter of 2020 to 2nd Quarter of 2020:** The exogenous shock of the Pandemic (COVID-19). Here, the convergence with the official dating is total, capturing the abrupt shutdown of economic activity.

Technical Verdict: The Signal Beyond the Model

The rawness of this exercise's results forces us to obliterate the "imminent economic recovery" narratives that, unfortunately, are often uttered by colleagues and economic analysts. When faced with the behavior of these regimes, what emerges is not a series of virtuous economic cycles, but a complex structural condition.

1. The Inertia Trap (The Weight of p_{11})

The 53.3% probability of remaining in a recessive regime is not just a statistical signal; it is the quantification of the system's inability to break away from its own downward trajectory. In Brazil, crises have "memory." Unlike economies that use collapse to clear inefficiencies and re-emerge, the model suggests that our recession is "mitochondrial"—it is already in our "DNA." The inertia is so strong that there is a self-feeding process of contraction, requiring shocks of disproportionate magnitude for any change of state to occur.

2. The Fragility of Expansion (The Weight of σ^2)

The most revealing data point, however, lies in the expansion regime. With a noise value of 1.1, which exceeds the growth rate itself (0.79%), the Brazilian expansion reveals itself as a state of absolute fragility, where the noise subjugates the growth signal. This means that, even in times of growth, the environment is so volatile that, in the face of any conjunctural interference, the gain will be marginal, if any.

3. The Cost of Linear Illusion

Considering the Brazilian economy as a linear system is a strategic error that no decision-maker can afford to neglect. Ignoring the non-linearity and the regime ruptures presented in this exercise is like "operating blindfolded." To be clear: the objective of this exercise was to raise questions and show that the promise of a virtuous economic cycle is extremely fragile.